Price-Only Long/Short M5 Strategy Study: Live-Trading Rejection Report

RegimeForgeEA Research

2026-07-11

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Executive conclusion

This report records a price-only, long/short M5 candidate study designed to meet stricter live-trading standards. It removes all volume inputs, uses next-bar bid/ask execution, explicit ATR stop-loss and take-profit exits, trailing stops, maximum holding periods, and account-level risk locks.

None of the four pre-defined candidates passed training. No candidate was therefore allowed to proceed to 2024 validation or the 2025 final holdout. This is a rejection report. The correct action is to keep the new price-only candidate out of the MQL5 execution path until a future candidate passes the same gates.

1. Reason for the revised study

Gold volume data are broker-dependent and inconsistent. Exchange activity, broker tick volume, CFD internalization, and session coverage can produce very different volume series. The earlier volume-confirmed proxy candidate is not an acceptable live-trading specification under the trader review note.

The revised study removes `volume`, `tick_volume`, `real_volume`, aggregate trades, and all order-flow fields from its signal definition. It uses only price-derived OHLC features and completed higher-timeframe trend information.

2. Live-trading requirements encoded in the study

The following requirements were applied before any result was inspected:

1. Both long and short entries must be possible.
2. Entries must use completed M5 signals and execute at the next bar's bid/ask.
3. Every position must have an ATR stop-loss and ATR take-profit level.
4. A trailing stop and a maximum holding time must be active.
5. Risk sizing must use a fixed percentage of equity rather than a fixed lot.
6. A daily loss lock and a peak-drawdown entry lock must be active.
7. Training, validation, and final holdout periods must remain separated.
8. No volume-based signal or filter may be used.

3. Price-only factor

Let C_t be the completed M5 close. For lookback k, define:

$$r_k(t) = \frac{C_t}{C_{t-k}} - 1$$

Let $q_L(t)$ and $q_U(t)$ be trailing lower and upper return quantiles, shifted one bar so that the signal bar is excluded:

$$q_L(t) = Q_q(\{r_k(s) : s < t\}), \qquad q_U(t) = Q_{1-q}(\{r_k(s) : s < t\})$$

Let the trend state be calculated from the prior completed H1 bar:

$$T_{up}(t) = EMA_{fast}^{H1}(t-1) > EMA_{slow}^{H1}(t-1)$$

$$T_{down}(t) = EMA_{fast}^{H1}(t-1) < EMA_{slow}^{H1}(t-1)$$

The symmetric signals are:

$$Long(t) = T_{up}(t) \wedge r_k(t) \leq q_L(t)$$

$$Short(t) = T_{down}(t) \wedge r_k(t) \geq q_U(t)$$

A quantile-cross condition prevents repeatedly entering while the same return remains at an extreme. The signal is still a trend-aligned pullback/reversal idea, but it is entirely price-derived.

4. Protective exits and sizing

For entry price E and signal-bar ATR A, long positions use:

$$SL = E - m_s A, \qquad TP = E + m_t A$$

Short positions use:

$$SL = E + m_s A, \qquad TP = E - m_t A$$

The engine resolves an OHLC bar touching both stop and target as stop-first, which is a conservative convention. A trailing stop updates after bar close. The position is also closed at the maximum configured holding time.

Position size is calculated from 0.25% equity risk and the initial stop distance. New entries are blocked when the daily equity loss reaches 2% or when the peak-equity drawdown reaches 20%.

5. Candidate universe

The universe was intentionally small to avoid broad parameter mining. All candidates are M5 execution models with symmetric long and short rules:

Candidate	H1 trend	Return quantile	Lookback	Stop / target	Trail	Maximum hold
PR01	EMA20 / EMA50	20%	3 bars	1.5 / 2.0 ATR	1.2 ATR	24 bars
PR02	EMA20 / EMA50	10%	3 bars	1.5 / 2.0 ATR	1.2 ATR	24 bars
PR03	EMA20 / EMA50	20%	3 bars	2.0 / 3.0 ATR	1.5 ATR	48 bars
PR04	EMA20 / EMA100	20%	6 bars	2.0 / 3.0 ATR	1.5 ATR	48 bars

6. Data and cost assumptions

The public study uses weekday-only PAXGUSDT M5 bars for 2021-2025 as a gold-linked proxy. This is not broker-native XAUUSD data. A fixed 0.35 USD round-trip spread is applied. Long entry uses ask-open and short entry uses bid-open. Long exits use bid prices and short exits use ask prices.

This study excludes commission, swap, slippage, latency, partial fills, broker stops-level restrictions, and real bid/ask historical behavior. Consequently, passing this study would only authorize an XAUUSD broker-native test; it would not authorize live deployment.

7. Time split and gates

The data split is fixed as follows:

Stage	Period	Purpose
Training	2021-2023	Candidate gate and ranking
Validation	2024	Independent selection check
Holdout	2025	Final one-time evaluation

Training requires at least 150 trades, PF >= 1.10, positive return, and maximum drawdown <= 15%. Validation requires at least 50 trades and the same PF, positive-return, and drawdown requirements. The holdout remains unread unless a candidate passes both earlier stages.

8. Training results

Candidate	Return	Maximum drawdown	Trades	Win rate	Profit factor
PR01 H1 q20 r3, 1.5/2.0 ATR, 24 bars	-16.58%	20.10%	2,871	36.29%	0.93

Candidate	Return	Maximum drawdown	Trades	Win rate	Profit factor
PR02 H1 q10 r3, 1.5/2.0 ATR, 24 bars	-0.10%	20.06%	3,740	37.83%	1.00
PR03 H1 q20 r3, 2.0/3.0 ATR, 48 bars	-18.68%	20.15%	2,322	35.57%	0.88
PR04 H1 slow q20 r6, 2.0/3.0 ATR, 48 bars	-19.74%	20.02%	2,255	35.70%	0.87

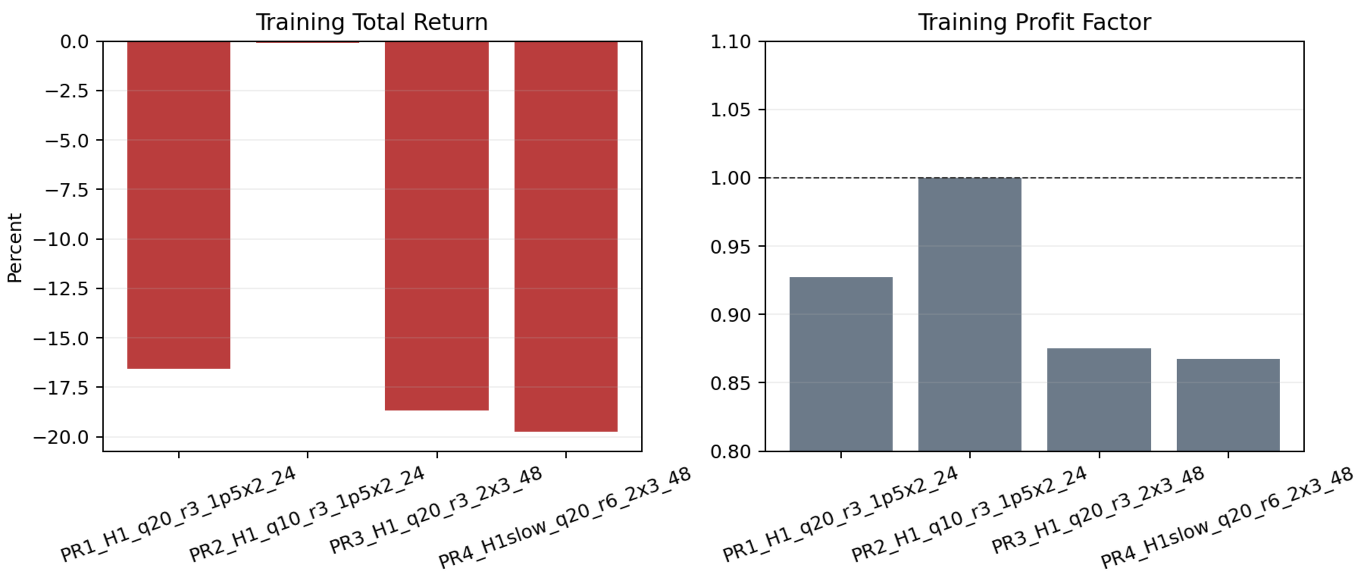

Figure 1: Training return and profit factor

9. Interpretation of the negative result

The closest candidate, PR02, is approximately breakeven before a full broker cost model, with PF 1.00 and a drawdown lock reached near 20%. This is not a tradable edge. The other candidates lose materially in training.

The negative result is informative. Once volume is removed and symmetric short trading is enforced, the simple higher-timeframe return-extreme rule does not have enough edge to pay for stops, targets, spread, and realistic risk control. Adding it to the EA would create the appearance of a live-ready strategy while contradicting the evidence.

10. Implications for the MQL5 EA

The existing volume-based EA must not be described as compliant with this new volume-free live-strategy standard. The price-only `price_reversal` signal is available in the Python engine for continued research, but no price-only MQL5 EA will be activated from this study because every candidate failed the first gate.

The safe operational state is:

- Keep `InpEnableNewEntries=false` for any research EA by default.
- Do not substitute volume with broker tick volume merely to preserve the old factor; that would violate the trader review requirement.
- Do not add a short-selling feature to the MQL5 EA until a symmetric long/short candidate passes the stated research protocol.

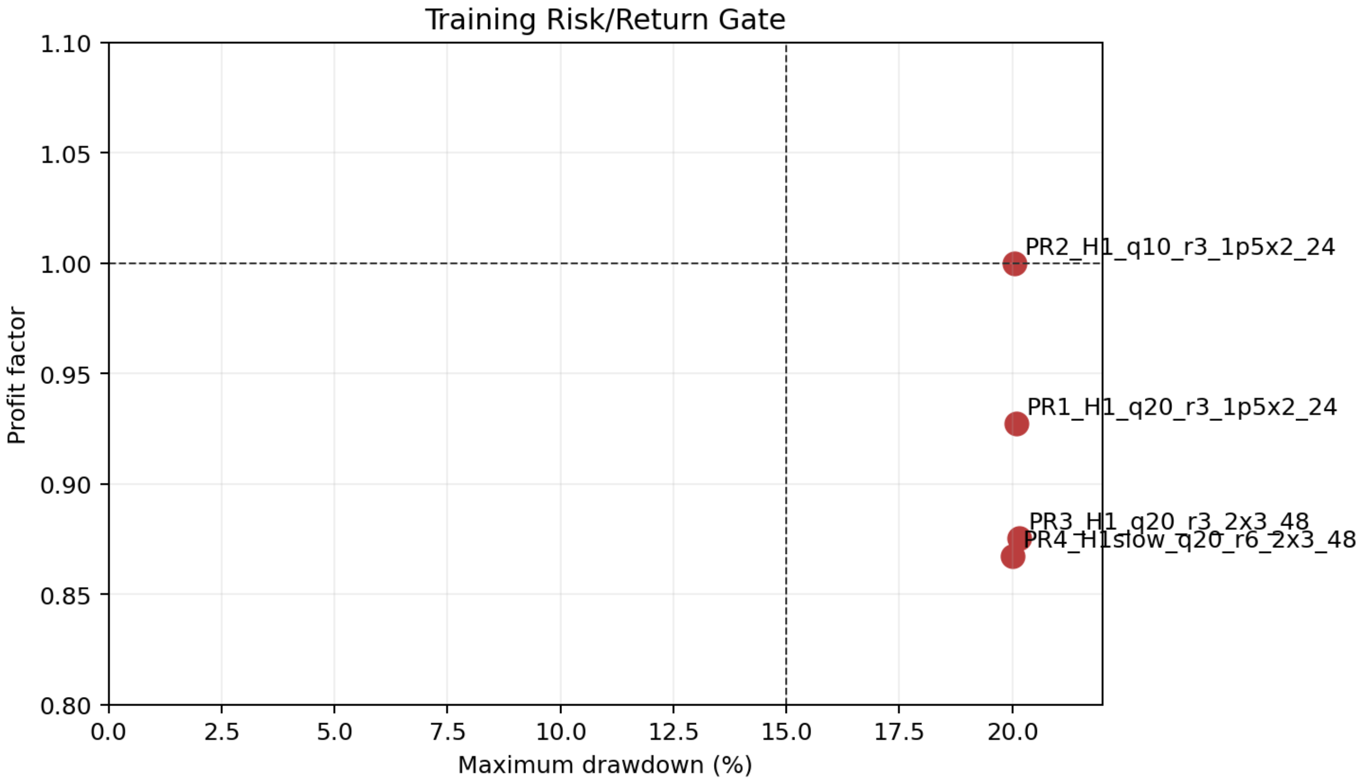

Figure 2: Training risk-return gate

11. Next research directions

The next candidate family should not be another small variation of the same return-quantile rule. More defensible price-only directions include:

1. Session-aware opening-range breakout with ATR stop/target and both sides.
2. Multi-timeframe trend continuation after volatility compression, without volume confirmation.
3. Price-action mean reversion only in statistically identified range regimes.
4. Separate long and short strategy families, each with independent evidence, rather than forcing symmetry where market microstructure is asymmetric.

Each family must be pre-defined, tested with protective exits, and subjected to the same train/validation/holdout gate before implementation.

12. Reproduction

```
python scripts/research_price_only_live_candidates.py \
  data/derived/PAXGUSDT_5m_2021_2025_weekdays.csv \
  --output-json outputs/price_only_live_candidates.json \
  --report reports/Price_Only_Live_Candidate_Research.md

python scripts/generate_price_only_research_charts.py \
  outputs/price_only_live_candidates.json \
  --assets reports/assets
```

13. Final decision

No price-only candidate in this study is approved for MQL5 integration, validation, or holdout testing. This preserves the live-trading research standard: a rejected strategy remains rejected, even when the requested design features are present.